

Eigenfaces for Recognition

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Introduction

Face recognition is a challenging problem in computer vision due to variations in lighting, facial expressions, and pose. The Eigenfaces method addresses this problem by applying Principal Component Analysis (PCA) to facial images, constructing a low-dimensional subspace known as *face space*. In this work, we implement the Eigenfaces algorithm from scratch using the Olivetti faces dataset, generate eigenfaces through PCA, and perform recognition by comparing projected face representations. We further analyze reconstruction quality and study how the number of eigenfaces affects recognition accuracy and dimensionality reduction.



Dataset Description

The experiments are performed on the Olivetti faces dataset, which contains 400 grayscale images of 40 individuals, with 10 images per person. Each image has resolution 64×64 , corresponding to a 4096-dimensional vector after flattening.

The dataset was collected under controlled conditions, with aligned faces, uniform backgrounds, and limited variations in lighting and pose. These properties make it well suited for evaluating the Eigenfaces method while still containing enough variation in expression and appearance to test recognition performance.

For training and evaluation, 7 images per subject are used for training and 3 for testing, giving:

280 training images 120 test images

Sample Images from the Dataset

Sample face from each of 32 subjects



Examples of facial images from the Olivetti dataset under controlled conditions.

Mathematical Background: Notation

Symbol	Meaning	Shape
Γ_i	Face image vector	N^2
Ψ	Mean face	N^2
Φ_i	Centered face	N^2
A	Data matrix	$N^2 \times M$
C	Covariance matrix	$N^2 \times N^2$
L	Reduced matrix	$M \times M$
u_l	Eigenface	N^2
Ω	Face-space vector	M'
ε_k^2	Distance to class	Scalar

Notation used throughout the Eigenfaces framework.

Image Representation

Each grayscale image is reshaped into a vector in a high-dimensional space.

For an image of size $H \times W$:

$$N^2 = H \times W$$

For the Olivetti dataset:

$$64 \times 64 = 4096$$

Each image is represented as:

$$\Gamma_i \in \mathbb{R}^{N^2}$$

Thus, facial images can be treated as points in a vector space, enabling the use of PCA for dimensionality reduction.

Mean Face

The dataset is first centered by computing the mean face:

$$\Psi = \frac{1}{M} \sum_{i=1}^M \Gamma_i$$

Each image is then mean-centered:

$$\Phi_i = \Gamma_i - \Psi$$

Centering removes global intensity bias and allows PCA to capture directions of maximum variation.

Covariance Matrix and PCA

PCA finds directions of maximum variance using the covariance matrix:

$$C = \frac{1}{M}AA^T$$

where

$$A = [\Phi_1, \Phi_2, \dots, \Phi_M]$$

Eigenvectors of C define the principal components, while eigenvalues measure the variance captured.

Direct computation is expensive since:

$$C \in \mathbb{R}^{N^2 \times N^2}$$

The $M \times M$ Covariance Trick

Instead of diagonalizing C , we compute eigenvectors of:

$$L = A^\top A$$

where

$$L \in \mathbb{R}^{M \times M}$$

If

$$A^\top A v_l = \mu_l v_l$$

then

$$u_l = A v_l$$

are eigenvectors of the covariance matrix.

Since $M \ll N^2$, this greatly reduces computational cost.

Eigenfaces and Face Space

The principal eigenvectors obtained from PCA are called *eigenfaces*.

Projection into face space:

$$\omega_k = u_k^\top \Phi$$

Face-space representation:

$$\Omega = [\omega_1, \omega_2, \dots, \omega_{M'}]^\top$$

Reconstruction from face space:

$$\Phi_f = \sum_{k=1}^{M'} \omega_k u_k$$

$$\Gamma \approx \Psi + \Phi_f$$

Reconstruction from Face Space

After projection into face space, a centered image can be approximately reconstructed using the retained eigenfaces.

For

$$\Phi = \Gamma - \Psi$$

the reconstruction is:

$$\hat{\Phi} = \sum_{k=1}^{M'} \omega_k u_k = U U^T \Phi$$

where

$$\omega_k = u_k^T \Phi$$

and

$$U = [u_1, u_2, \dots, u_{M'}]$$

The reconstructed image is:

$$\hat{\Gamma} = \Psi + \hat{\Phi} = \Psi + U U^T (\Gamma - \Psi)$$

As more eigenfaces are retained, reconstruction quality improves and finer facial details are recovered.

Reconstruction Error and Energy Preservation

The reconstruction error measures information lost during dimensionality reduction:

$$\varepsilon^2 = \|\Phi - \hat{\Phi}\|^2$$

Using orthonormal eigenfaces:

$$\varepsilon^2 = \sum_{l=M'+1}^M \omega_l^2$$

Thus, the error equals the energy contained in discarded eigen-directions.

By Parseval's theorem:

$$\|\Phi\|^2 = \sum_{l=1}^M \omega_l^2$$

Retained energy fraction:

$$\rho(M') = \frac{\sum_{l=1}^{M'} \omega_l^2}{\sum_{l=1}^M \omega_l^2}$$

This explains why relatively few eigenfaces preserve most important facial information.

The Eigenfaces method consists of two stages:

① Training Phase

- Construct the face space using PCA
- Compute eigenfaces from training images
- Store projected representations

② Recognition Phase

- Project a new image into face space
- Compare against known classes
- Decide whether the image is known, unknown, or not a face

Training Phase: Data Preparation

Each grayscale image is reshaped into a vector:

$$\Gamma_i \in \mathbb{R}^{N^2}$$

The training dataset is represented as:

$$\{\Gamma_1, \Gamma_2, \dots, \Gamma_M\}$$

The mean face is computed:

$$\Psi = \frac{1}{M} \sum_{i=1}^M \Gamma_i$$

Each image is then centered:

$$\Phi_i = \Gamma_i - \Psi$$

The centered images form the data matrix:

$$A = [\Phi_1, \Phi_2, \dots, \Phi_M]$$

Training Phase: Computing Eigenfaces

Instead of diagonalizing the large covariance matrix directly, we compute:

$$L = A^T A$$

and solve:

$$Lv_l = \mu_l v_l$$

The eigenvectors are mapped back into image space:

$$u_l = Av_l$$

and normalized:

$$u_l \leftarrow \frac{u_l}{\|u_l\|}$$

These vectors u_l are the *eigenfaces*.

The top M' eigenfaces form the basis matrix:

$$U = [u_1, u_2, \dots, u_{M'}]$$

Training Phase: Projection into Face Space

Each centered training image is projected into face space:

$$\Omega_i = U^\top (\Gamma_i - \Psi)$$

For each class k , a centroid is computed:

$$\Omega_k = \frac{1}{n_k} \sum_{i \in \text{class } k} \Omega_i$$

These projected vectors provide compact facial representations used during recognition.

Recognition Phase

Given a new image Γ , we first center it:

$$\Phi = \Gamma - \Psi$$

Projection into face space:

$$\Omega = U^T \Phi$$

Distance to each class centroid:

$$\varepsilon_k = \|\Omega - \Omega_k\|$$

The nearest centroid gives the closest facial match.

Reconstruction from face space:

$$\Phi_f = U\Omega$$

Reconstruction error:

$$\varepsilon = \|\Phi - \Phi_f\|$$

Recognition Decision Rule

Classification is determined using two thresholds:

$$\theta_{\varepsilon} \quad \theta_{\xi}$$

- If

$$\varepsilon < \theta_{\xi} \quad \text{and} \quad \varepsilon_k < \theta_{\varepsilon}$$

the image is recognized as class k

- If

$$\varepsilon < \theta_{\xi} \quad \text{and} \quad \varepsilon_k \geq \theta_{\varepsilon}$$

the image is treated as an unknown face

- If

$$\varepsilon \geq \theta_{\xi}$$

the input is classified as not a face

Results: Mean Face

The mean face represents the average structure shared across all training images.

It appears blurred because individual facial variations are averaged together, while common facial features remain visible.

Mean face Ψ

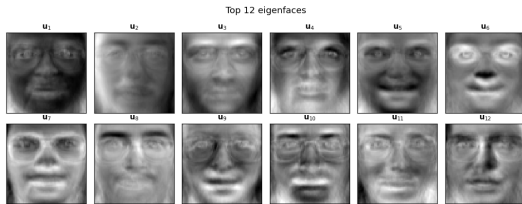


Results: Eigenfaces

The principal eigenvectors are called *eigenfaces*.

Leading eigenfaces capture:

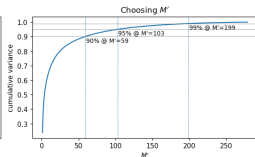
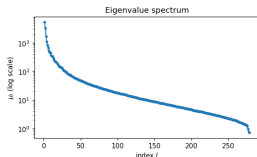
- Lighting variation
- Facial structure
- Expression changes



Eigenvalue Spectrum and Dimensionality

The eigenvalue spectrum decays rapidly, showing that most variance is captured by relatively few components.

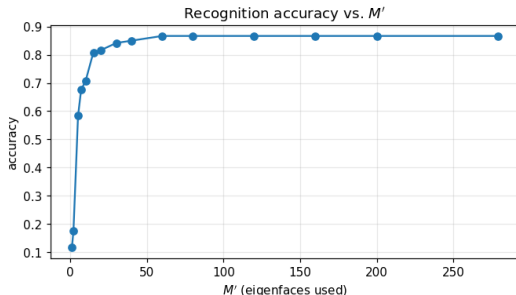
- 40–80 eigenfaces are often sufficient
- Higher components contribute little additional information
- Recognition does not require perfect reconstruction



Recognition Accuracy vs. Number of Eigenfaces

Recognition accuracy improves as more eigenfaces are retained.

- Small M' gives poor representation
- Accuracy rises rapidly initially
- Performance saturates near 85%–87%



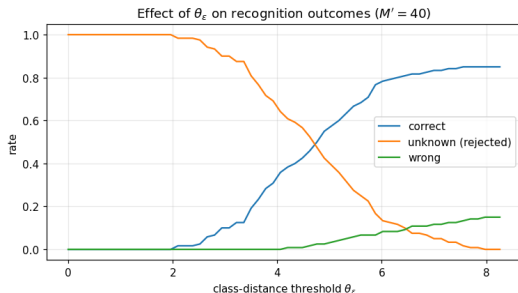
Most discriminative information lies in the leading principal components.

Threshold Analysis

Changing the threshold affects the trade-off between:

- Correct recognition
- Rejection of unknown inputs
- Misclassification

Lower thresholds reduce false positives but reject more inputs.



Reconstruction Quality

With few eigenfaces, reconstructions retain only coarse structure.

As more eigenfaces are included, finer facial details are restored.

This demonstrates PCA's ability to provide compact yet expressive representations.



Face vs. Non-Face Detection

Reconstruction error helps distinguish faces from non-face inputs.

- Genuine faces project close to face space
- Non-face inputs produce larger reconstruction error

This allows rejection of unlikely facial inputs.

training face
input



projection, $\epsilon=3.91$



held-out face
input



projection, $\epsilon=5.23$



random noise
input



projection, $\epsilon=18.58$



Limitations

Major limitations include:

- Sensitivity to lighting
- Dependence on aligned faces
- Difficulty modeling nonlinear variations
- Sensitivity to noise and outliers

Lighting variation can dominate identity-related information in PCA.



Conclusion

In this work, we implemented the Eigenfaces method for face recognition from first principles using Principal Component Analysis and linear algebra techniques. By projecting facial images into a lower-dimensional face space, the method achieves compact representation while preserving the most important facial variations required for recognition. Our experiments demonstrate that accurate classification can be achieved using only a relatively small number of eigenfaces, with performance improving as the subspace dimension increases before eventually saturating. Reconstruction and threshold analyses further illustrate both the interpretability and limitations of the method. Although Eigenfaces performs best under controlled conditions and remains sensitive to variations such as lighting and pose, it continues to serve as an important foundational approach in computer vision and pattern recognition.